SochX: Where Ideas Go Live

Mukammal Project Architecture, Tech Stack aur Implementation Guide

Project Type: Full-Stack Web Application (MERN / PERN / MySQL)

Frontend: React 19 + Vite

Backend: Node.js + Express.js

Database: MySQL (XAMPP / MariaDB)

1. Project ka Maqsad (Overview & Problem Solved)

SochX ek mukammal aur modern Full-Stack platform hai jo students, innovators aur professionals ko apne creative ideas, startup proposals aur social/tech issues ke solutions share karne ki sahulat deta hai.

- **Crowdsourced Innovation:** Log apne ideas categorize kar ke post kar sakte hain (jaise Technology, Education, Health, Business, Social, Sustainability).
- **Community Engagement:** Doosre users in ideas ko parh sakte hain aur **Upvote** de kar unhein promote kar sakte hain.
- **Strict Moderation:** Admin panel har idea ko verify karta hai aur misuse ya low-quality content ko rokne ke liye **Approve** ya **Reject** karta hai.
- **Role-Based Security:** Admins aur normal users ke darmiyan wazeh farq rakha gaya hai taake unauthorized access na ho sake.

2. System Architecture & Request Lifecycle

Project ko 3-tier architecture par banaya gaya hai jisme Client, Server aur Database alag alag responsibilities sambhalte hain:

```
[ React Frontend (Client - Port 5173) ]
         │  HTTP Request (Axios) + JWT Token
         ▼
[ Express.js Server (Backend API - Port 5000) ]
         │  CORS + JSON Body Parser Middleware
         ▼
[ Authentication Middleware (JWT Verification) ]
         │  Valid Token -> req.user attach
         ▼
[ Controller (e.g. ideaController / authController) ]
         │  Business Rules + ACID Transaction
         ▼
[ MySQL Connection Pool (mysql2/promise) ]
         │  Parameterized Prepared SQL Queries
         ▼
[ Database: sochx_db (Tables: users, ideas, idea_upvotes, comments) ]
```

3. Kon Si Cheezein Use Hui Hain Aur Kyun Use Hui Hain?

A. Frontend (React 19 + Vite)

Tool / Library	Folder / File	Istemal ki Wajah (Purpose & Reason)
React 19	`src/`	Fast, responsive aur component-driven UI banane ke liye. Page reload hue baghair UI live update hoti hai.
Vite	`vite.config.js`	Modern build tool aur development server. Iski Hot Module Replacement (HMR) bohot tez hai.
React Router Dom (v7)	`src/App.jsx`	Single Page Application (SPA) routing ke liye. `/`, `/ideas`, `/submit-idea`, `/profile`, `/admin` par navigate karne ke liye.
Axios	`src/services/api.js`	Backend REST API se communication ke liye. Isme centralized interceptors hain jo JWT token khud attach karte hain.
Lucide React	`src/components/`	Modern, lightweight icons (Thumbs up, Trash, Search, Filters, User profile icons) ke liye.
Tailwind CSS	`tailwind.config.js`	Responsive layout, badges, card animations aur professional dark/light theme ke liye.

B. Backend (Node.js + Express + MySQL)

Package / Tool	Folder / File	Istemal ki Wajah (Purpose & Reason)
Express.js	`backend/server.js`	REST API server create karne aur HTTP requests (GET, POST, PUT, DELETE) handle karne ke liye.
MySQL2 (Promise)	`backend/config/db.js`	Asynchronous queries aur Connection Pool manage karne ke liye taake server heavy load par crash na ho.
JSON Web Token (JWT)	`backend/middleware/`	Stateless authentication ke liye. Login hone par client ko secure token diya jata hai jo user ki identity verify karta hai.
Bcrypt.js	`authController.js`	Passwords ko secure salt (10 rounds) ke sath hash banane ke liye taake database me plain passwords na jayein.
CORS	`backend/server.js`	Port 5173 (Frontend) ko Port 5000 (Backend) ke resources access karne ki ijazat dene ke liye.
Dotenv	`backend/.env`	Sensitive environment variables (Database password, Port, JWT secret) ko code se alag mehfooz rakhne ke liye.

4. Database Design & Tables Structure (sochx_db)

Database me 4 relational tables banayi gayi hain jisme foreign keys aur indexing shamil hai:

1. **users Table:** Users ke accounts, role (`'admin'` ya `'user'`), hashed passwords, role badges aur bio ko store karti hai.
2. **ideas Table:** Idea ka title, problem description, proposed solution, category, submitter info, status (`'Pending'` , `'Approved'` , `'Rejected'`), aur upvotes count rakhti hai.
3. **idea_upvotes Table:** Isme `UNIQUE KEY (ideaId, userId)` lagaya gaya hai. Yeh database level par guarantee karta hai ke **1 user sirf 1 baar upvote kar sake**.
4. **comments Table:** Ideas par hone wali discussion aur feedback messages ko user information ke sath link karti hai.

5. Step-by-Step Kaam Kaise Chalta Hai (Workflow)

> **Step 1: User Registration & Authentication**
> User apna naam, email, username aur password enter karta hai. Backend me password `bcrypt.hash()` ke zariye encrypt hota hai aur MySQL me save hota hai. Login hone par JWT token generate ho kar frontend ko milta hai jo `localStorage` me save hota hai.

Step 2: Ideas Feed & Live Search
User `/ideas` page par categories (Technology, Health, Education, Business, Social, Sustainability) select kar sakta hai. Search bar me text likhte hi backend dynamic SQL query chalata hai: `WHERE (title LIKE ? OR problemDescription LIKE ?) AND category = ?`

Step 3: New Idea Submission
User `/submit-idea` par apna naya concept submit karta hai. Idea default taur par **"Pending"** status me chala jata hai taake admin approval ke baghair feed me spam na aaye.

Step 4: Upvoting System (Concurrency Safe)
Jab user Thumbs Up click karta hai, to backend SQL Transaction (`connection.beginTransaction()` aur `FOR UPDATE`) use karta hai. Agar pehle se voted hai to vote remove ho jata hai (unlike), warna vote insert hota hai aur count barh jata hai.

Step 5: Admin Moderation Panel (Protected)
Sirf Admin role wale users `/admin` access kar sakte hain. Frontend ka `AdminRoute` component unko protect karta hai. Admin wahan se live metrics dekh sakta hai, ideas ko Approve/Reject kar sakta hai aur users ko manage kar sakta hai.

6. Project ko Run Karne ka Mukammal Tarika

1. XAMPP Server Setup

- XAMPP Control Panel open karein aur **Apache** aur **MySQL** start karein.
- Browser me `http://localhost/phpmyadmin` open karein.
- Database ka naam `sochx_db` rakhein aur `sochx_db.sql` file ko import karein.

2. Backend Start Karna

```
cd backend
npm install
npm run dev
# Server http://localhost:5000 par start ho jayega
```

3. Frontend Start Karna

```
# Project root directory me
npm install
npm run dev
# Frontend http://localhost:5173 par live ho jayega
```

7. Testing ke Liye Pre-configured Accounts

Role	Email Address	Password	Access Level
Admin	`arqam@sochx.com`	`Admin@123`	Admin Dashboard, Approve/Reject Ideas, User Management
Admin	`faryal@sochx.com`	`Lead@123`	Co-Founder / Admin Permissions
Standard User	`hamza@sochx.com`	`User@123`	Submit Ideas, Upvote, Profile Edit, Public Feed